

9	(a) (i)	<i>either</i> probability of decay (of a nucleus) per unit time <i>or</i> $\lambda = (-)(dN/dt) / N$ $(-)(dN/dt)$ and N explained	M1 A1 (M1) (A1)	[2]
	(ii)	in time $t_{1/2}$, number of nuclei changes from N_0 to $\frac{1}{2}N_0$ $\frac{1}{2} = \exp(-\lambda t_{1/2})$ <i>or</i> $2 = \exp(\lambda t_{1/2})$ $\ln(\frac{1}{2}) = -\lambda t_{1/2}$ and $\ln(\frac{1}{2}) = -0.693$ <i>or</i> $\ln 2 = \lambda t_{1/2}$ and $\ln 2 = 0.693$ $0.693 = \lambda t_{1/2}$	B1 B1 B1 A0	
	(b)	$228 = 538 \exp(-8\lambda)$ $\lambda = 0.107 \text{ (hours}^{-1}\text{)}$ $t_{1/2} = 6.5 \text{ hours (do not allow 3 or more SF)}$	C1 C1 A1	[3]
	(c)	e.g. random nature of decay background radiation daughter product is radioactive <i>(any two sensible suggestions, 1 each)</i>	B2	

Page 6	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2012	9702	41

Section B

10. (a) light dependent resistor (allow LDR)

B1

[1]